

# DEADLOCK DETECTION ALGORITHM

## C PROGRAM:

```
#include <stdio.h>

int main()
{
    int allocation[10][10], request[10][10];
    int available[10];
    int work[10];
    int finish[10];
    int n, m;
    int i, j, k;
    int found;

    printf("Enter Number of Processes: ");
    scanf("%d", &n);

    printf("Enter Number of Resource Types: ");
    scanf("%d", &m);

    printf("\nEnter Allocation Matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < m; j++)
        {
            scanf("%d", &allocation[i][j]);
        }
    }

    printf("\nEnter Request Matrix:\n");

    for(i = 0; i < n; i++)
```

```
{  
for(j = 0; j < m; j++)  
{  
scanf("%d", &request[i][j]);  
}  
}  
printf("\nEnter Available Resources:\n");
```

```
for(i = 0; i < m; i++)  
{  
scanf("%d", &available[i]);  
work[i] = available[i];  
}
```

```
for(i = 0; i < n; i++)  
{  
finish[i] = 0;  
}  
do  
{  
found = 0;  
for(i = 0; i < n; i++)  
{  
if(finish[i] == 0)  
{  
for(j = 0; j < m; j++)  
{  
if(request[i][j] > work[j])  
break;
```

```
}  
if(j == m)  
{  
    for(k = 0; k < m; k++)  
    {  
        work[k] += allocation[i][k]  
    }  
    finish[i] = 1;  
    found = 1;  
}  
}  
}  
} while(found);  
found = 0;  
printf("\nDeadlocked Processes:\n");
```

```
for(i = 0; i < n; i++)  
{  
    if(finish[i] == 0)  
    {  
        printf("P%d ", i);  
        found = 1;  
    }  
}  
if(found == 0)  
{  
    printf("No Deadlock Detected");  
}
```

```
printf("\n");
```

```
return 0; }
```

## SHELL SCRIPTING:

```
#!/bin/bash
```

```
echo "Enter number of processes:"
```

```
read n
```

```
echo "Enter deadlocked process numbers (if any):"
```

```
read processes
```

```
if [ -z "$processes" ]
```

```
then
```

```
echo "No Deadlock Detected"
```

```
else
```

```
echo "Deadlocked Processes:"
```

```
echo "$processes"
```

```
fi
```

## OUTPUT:

```
(kali㉿ Anbuchelvan)-[~]  
$ nano deadlock.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ chmod +x deadlock.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ bash deadlock.sh
```

```
Enter number of processes:
```

```
2
```

```
Enter deadlocked process numbers (if any):
```

```
1
```

```
Deadlocked Processes:
```

```
1
```